

Apparent Instantaneity in T0 Theory

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Field equation $\nabla^2 m = 4\pi G \rho m$ as “wave equation with finite propagation speed”: annotated – without a time derivative it is elliptic (Poisson/Helmholtz type); only the wave equation with $\partial^2 T / \partial t^2$ is hyperbolic.
- Prediction $\Delta t = \xi r / c$ for Bell tests: annotated – corresponds to $c / \xi = 7500 c$ and lies below the measured lower bound of $1.38 \times 10^4 c$.

Abstract

This work demonstrates that the apparent instantaneity in the T0 formalism arises from the notation of the local constraint condition $T \cdot E = 1$. Through analysis of the underlying field equations and hierarchical time scales, it is shown that T0 theory provides a causal description of quantum phenomena that is compatible with special relativity. The theory works with the single geometric parameter ξ ; SI quantities serve for unit conversion. The work extends the analysis to the complete duality between time, mass, energy, and length, and critically discusses the limits of interpretation in extreme situations.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

.1 Introduction: The Instantaneity Problem

Since the work of Einstein, Podolsky, and Rosen in the 1930s, physics has been occupied with a basic paradox: quantum mechanics appears to require instantaneous correlations between arbitrarily distant particles, which Einstein called “spooky action at a distance.” This apparent instantaneity manifests in various phenomena—from wave function collapse through Bell inequality violations to quantum entanglement.

The T0 formalism offers an alternative resolution to this paradox. The core idea is that the fundamental relationship between time and energy, expressed by the equation $T \cdot E = 1$, is often misunderstood. What appears at first glance to be an instantaneous coupling proves upon closer examination to be a local constraint condition that implies no action at a distance.

To understand this, we must distinguish between two fundamentally different types of physical relationships: local constraint conditions that apply at the same spatial point, and field equations that describe the propagation of disturbances through space. This distinction is the key to resolving the instantaneity paradox.

.2 Apparent Instantaneity in the T0 Formalism

The T0 equations appear to imply instantaneity at first glance, but this is refuted through detailed analysis of the field equations. The fundamental challenge is understanding how a theory based on the strict relationship $T \cdot E = 1$ can nonetheless respect causality. This

apparent paradox has its roots in a misunderstanding about the nature of mathematical constraint conditions in physics.

The Apparent Problem

The fundamental equations of the T0 formalism are:

$$T(\mathbf{x}, t) \cdot E(\mathbf{x}, t) = 1 \quad (1)$$

$$T = \frac{1}{m} \quad \text{where } \omega = \frac{mc^2}{\hbar}, \text{ so } T = \frac{\hbar}{E} \quad (2)$$

$$E = mc^2 \quad (3)$$

These equations suggest that a change in E requires an immediate adjustment of T . If we double the energy at a point, for example, the time field seems to have to halve instantaneously. This interpretation would indeed mean a violation of relativistic causality and stands in apparent contradiction to the fundamental principles of modern physics.

The confusion arises from the fact that these equations are often interpreted as dynamic relationships—as if a change in one quantity causes an instantaneous reaction in the other. This reading does not apply to the T0 equations; it is where the apparent paradoxes of quantum mechanics come from.

The Resolution: Field Equations Have Dynamics

The resolution of this paradox lies in recognizing that the T0 equations contain two different types of relationships: local constraint conditions and dynamic field equations. This distinction is fundamental to understanding why no real instantaneity occurs.

1. The complete field equation:

$$\nabla^2 m = 4\pi G \rho(\mathbf{x}, t) \cdot m \quad (4)$$

where $\rho(\mathbf{x}, t)$ is the mass density. This equation is *not* instantaneous but rather a wave equation with finite propagation speed $v \leq c$. (Note of 30 Sept. 2026: The equation contains no time derivative; it is elliptic (Poisson/Helmholtz type) and describes an instantaneously fixed field state, not propagation with $v \leq c$. "Second order in spatial derivatives" is not characteristic of wave propagation; that requires the second time derivative as in the full wave equation $\partial^2 T / \partial t^2 = c^2 \nabla^2 T + Q$. The same applies to the label " $v \leq c$ " attached to this equation in the figure.)

This field equation describes how disturbances in the mass field (and thus in the time field via $T = 1/m$) propagate through space. Crucially, this propagation occurs at finite speed, limited by the speed of light. The equation is second-order in spatial derivatives, which is characteristic of wave propagation. No information, no energy, and no effect can propagate faster than the speed of light.

2. The modified Schrödinger equation:

$$i \cdot T(\mathbf{x}, t) \frac{\partial \psi}{\partial t} = H_0 \psi + V_{T0} \psi \quad (5)$$

where $H_0 = -\frac{\hbar^2}{2m} \nabla^2$ is the free Hamiltonian and $V_{T0} = \hbar^2 \delta E(\mathbf{x}, t)$ is the T0-specific potential.

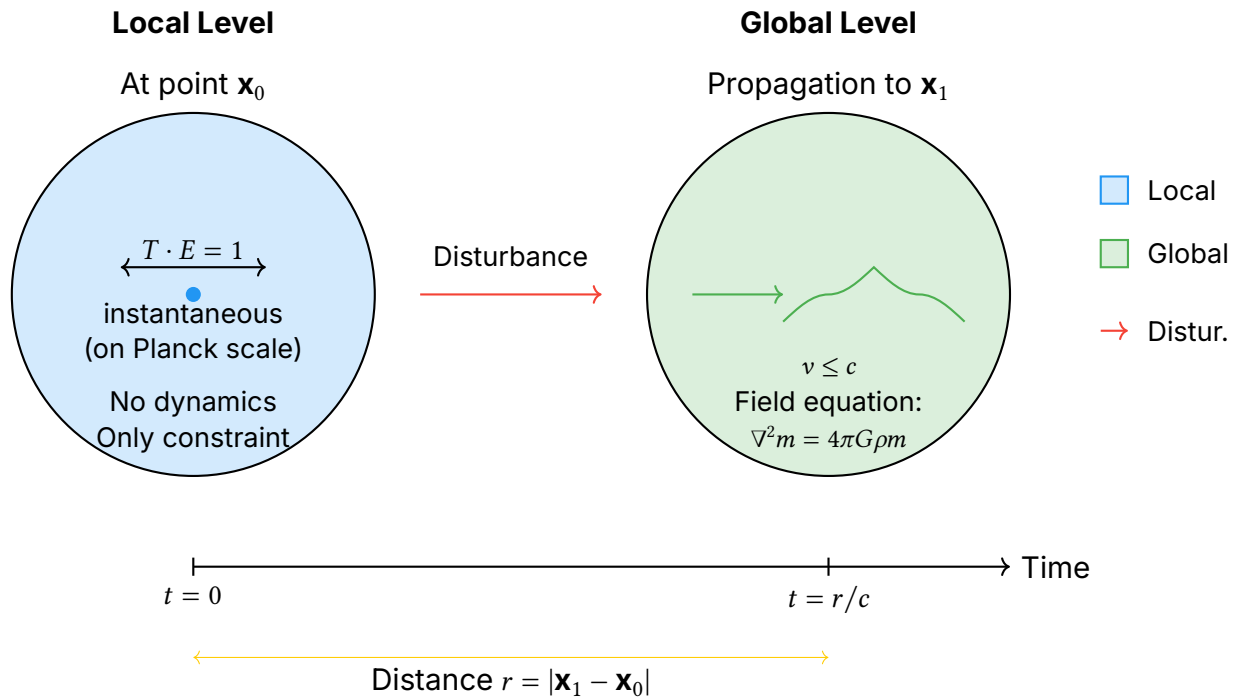
This modified Schrödinger equation explicitly shows the temporal evolution of the wave function under the influence of the time field. The presence of the time derivative $\partial/\partial t$ makes clear that this is a causal evolution, not an instantaneous adjustment. The wave function evolves continuously in time according to local field conditions.

.3 The Critical Insight: Local vs. Global Relations

The key to understanding lies in distinguishing between local and global physical relationships. This distinction is ubiquitous in physics but often not emphasized explicitly enough. The confusion between these two types of relationships is the source of many conceptual problems in quantum mechanics.

Visualization of Local vs. Global Relations

Local Constraint vs. Global Propagation



This diagram illustrates the fundamental difference between local and global processes. On the left, we see the local constraint condition $T \cdot E = 1$, which holds instantaneously (on the Planck time scale) at the same spatial point. On the right, we see the global propagation of a disturbance, which occurs at finite speed $v \leq c$ and requires time $t = r/c$ to bridge the distance r .

Local Constraint Condition

$$T(\mathbf{x}, t) \cdot E(\mathbf{x}, t) = 1 \quad [\text{AT THE SAME SPATIAL POINT}] \quad (6)$$

This is a local constraint condition—analogous to $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ in electrodynamics. It holds instantaneously at the same point but does not enforce instantaneous action at a distance.

To deepen this analogy: In electrodynamics, Gauss's law means that the divergence of the electric field at each point is proportional to the local charge density. This is not a statement about how changes propagate, but a condition that must be satisfied locally at each moment in time. When the charge density changes at a point, the electric field there adjusts immediately, but this change then propagates to other points at the speed of light.

The same applies to the T-E relationship in the T0 formalism. The equation $T \cdot E = 1$ is a local condition that must be satisfied at each spatial point at each moment. It does not describe how changes propagate, only the local relationship between the fields.

Causal Field Propagation

Change at $\mathbf{x}_1 \rightarrow$ Propagation with $v \leq c \rightarrow$ Effect at $\mathbf{x}_2$ (7)

$$\text{Time delay: } \Delta t = \frac{|\mathbf{x}_2 - \mathbf{x}_1|}{c} \quad (8)$$

The actual propagation of field changes follows the dynamic field equations. When the energy field changes at point $\mathbf{x}_1$, the time field there must immediately satisfy the constraint condition. However, this local change creates a disturbance in the field that propagates at finite speed.

The crucial point is that local adjustment and global propagation are two completely different processes. Local adjustment occurs on the Planck time scale and is practically instantaneous for all measurable purposes. Global propagation, however, is limited by the speed of light and can take considerable time over macroscopic distances.

.4 The Geometric Origin of T0 Parameters

A central aspect of T0 theory is that its parameter ξ is not empirically adjusted but derived from geometric principles. This distinguishes it from phenomenological approaches and makes its predictions testable.

Fundamental Geometric Derivation

T0 theory traces the physical parameters back to the geometry of three-dimensional space; integer structural choices are motivated settings, and SI anchors such as m_e serve for unit conversion. The central parameter is:

T0 Prediction

The universal parameter

$$\xi = \frac{4}{3} \times 10^{-4} \quad (9)$$

follows from purely geometric principles:

- Effective fractal dimension: $D_f^{\text{eff}} \approx 2.973$ (local: $D_f^{\text{space}} = 3 - \xi \approx 2.9999$)
- Ratio of characteristic scales to Planck length
- Topological properties of the quantum vacuum

This is *not* an empirical adjustment but a geometric prediction.

This is the core of the approach. While most physical theories contain free parameters that must be determined from experiments, ξ is derived here from the structure of space. The theory is therefore predictive and not merely descriptive.

The parameter ξ appears in various contexts and connects seemingly unrelated phenomena. It determines the strength of quantum corrections, the size of vacuum fluctuations, and the characteristic scales at which new physics appears. This universality supports treating ξ as a fundamental constant.

Experimental Confirmation

The geometric predictions of T0 theory are compared with precision measurements in the specialised documents of the corpus, without any parameter adjustment. Where the agreement holds, it supports the T0 approach.

That a parameter derived from geometry can be tested experimentally is the real test of the approach. If the agreement holds up, the structure of space itself determines the observed physical phenomena.

.5 Mathematical Specification of Field Dynamics

The mathematical structure of T0 field dynamics shows that the processes occur causally. This specification is needed to resolve the apparent paradoxes and to show that T0 theory is compatible with relativity.

Complete Wave Equation

T0 field dynamics follows the equation:

$$\frac{\partial^2 T}{\partial t^2} = c^2 \nabla^2 T + Q(T, E, \rho) \quad (10)$$

where the source function

$$Q(T, E, \rho) = -4\pi G\rho \cdot T \quad (11)$$

describes the self-interaction of the time field.

This wave equation carries the argument. It explicitly shows that the time field follows a hyperbolic differential equation characteristic of wave propagation at finite speed. The second derivatives with respect to time and space are in a fixed ratio given by the speed of light c . This guarantees that no information can be transmitted faster than light.

Example: Energy Change and Field Propagation

To illustrate the causal nature of field propagation, consider a concrete example:

$$t = 0 : \quad E(\mathbf{x}_0) \text{ changes} \quad (12)$$

$$\rightarrow T(\mathbf{x}_0) = \frac{1}{E(\mathbf{x}_0)} \quad [\text{local, constraint}] \quad (13)$$

$$\rightarrow \nabla^2 T \neq 0 \quad [\text{creates field disturbance}] \quad (14)$$

$$\rightarrow \text{Wave propagates with } v = c \quad (15)$$

$$t = \frac{r}{c} : \quad \text{Disturbance reaches point } \mathbf{x}_1 \quad (16)$$

This process clearly shows the hierarchy of events: local adjustment occurs immediately (on the Planck time scale), but propagation to distant points is limited by the speed of light.

.6 Green's Function and Causality

The Green's function is the mathematical tool that completely characterizes the causal structure of field propagation. It describes how a point disturbance propagates through the field and is thus fundamental to understanding causality in T0 theory.

The Green's function of the T0 field equation:

$$G(\mathbf{x}, \mathbf{x}', t - t') = \theta(t - t') \cdot \frac{\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))}{4\pi|\mathbf{x} - \mathbf{x}'|} \quad (17)$$

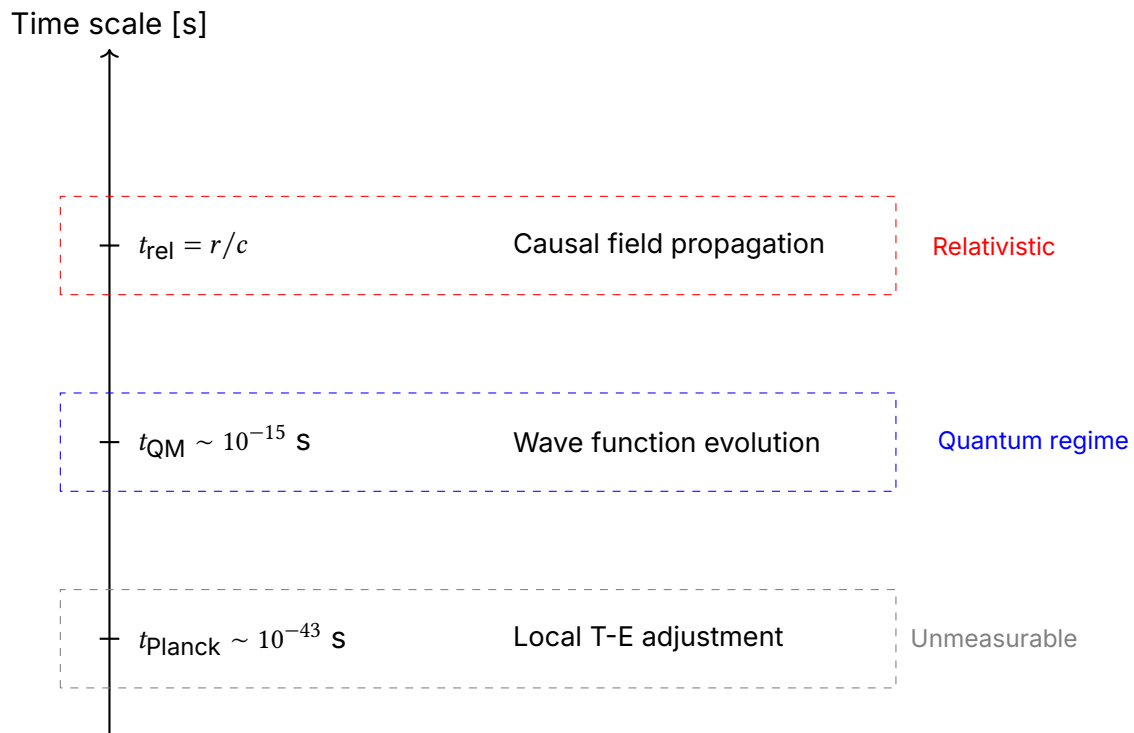
The components have the following meaning:

- $\theta(t - t')$: Heaviside function guarantees causality (effect after cause)
- δ function: encodes propagation at speed of light
- $1/4\pi r$: geometric factor for 3D propagation

Causality can be read off directly from the structure of this Green's function. The Heaviside function $\theta(t - t')$ is zero for $t < t'$, meaning no effect can occur before its cause. This is the mathematical implementation of the causality principle. The delta function $\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))$ is non-zero only when the distance equals c times the elapsed time—this describes a disturbance propagating exactly at the speed of light.

.7 The Hierarchy of Time Scales

Apparent instantaneity in quantum mechanics results from the extreme separation of different time scales. This hierarchy is fundamental to understanding why many quantum processes appear instantaneous even though they are not.



This hierarchy explains many seemingly paradoxical aspects of quantum mechanics. Processes on the Planck scale are so fast that they cannot be temporally resolved with any conceivable technology. For all practical purposes, they appear instantaneous. The

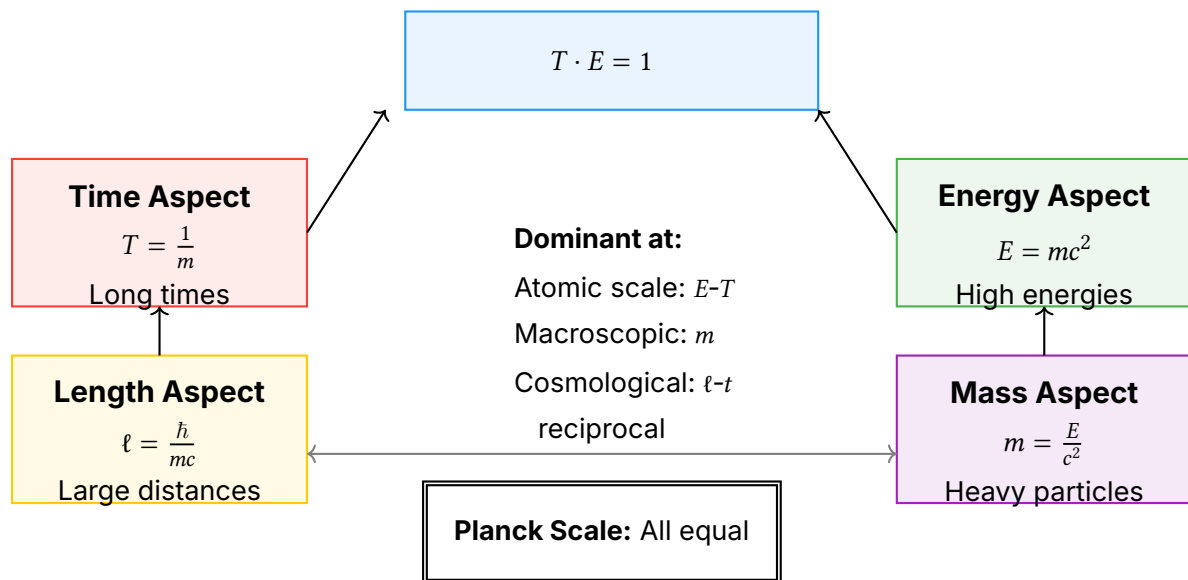
quantum scale is accessible to modern experiments but still extremely fast compared to macroscopic time scales. Finally, the relativistic scale determines propagation over macroscopic distances.

.8 The Complete Duality: Time, Mass, Energy, and Length

Beyond the time-mass duality, T0 theory describes a system of dualities linking time, mass, energy, and length. This extended perspective helps in understanding apparent instantaneity; the different physical quantities appear in it as aspects of the same underlying structure.

Visualization of Energy-Time Duality

The Fundamental Energy-Time Duality



Complementarity Principle:

The more precisely T is determined, the less precise E

$$\Delta T \cdot \Delta E \geq \frac{\hbar}{2}$$

This diagram shows the fundamental energy-time duality and its connections to mass and length. The central relationship $T \cdot E = 1$ connects all aspects. Depending on the scale considered, different aspects of this duality dominate, but all are linked by the fundamental relationships.

The Fundamental Equivalences

In the T0 formalism, the basic physical quantities are linked by the following relationships:

$$T \cdot E = 1 \quad (\text{Time-Energy duality}) \quad (18)$$

$$T = \frac{1}{m} \quad (\text{Time-Mass relation}) \quad (19)$$

$$E = mc^2 \quad (\text{Mass-Energy equivalence}) \quad (20)$$

$$\ell = \frac{\hbar}{mc} = \frac{\hbar}{E/c} \quad (\text{Length as energy}) \quad (21)$$

These relationships show that lengths can also be interpreted as energy scales. The Compton wavelength $\lambda_C = \hbar/(mc)$ is the paradigmatic example: it represents the characteristic length scale at which the quantum nature of a particle with mass m (or equivalently, energy $E = mc^2$) becomes manifest.

The Planck Scale as Universal Reference

All these dualities converge at the Planck scale:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (\text{Planck length}) \quad (22)$$

$$t_P = \sqrt{\frac{\hbar G}{c^5}} \quad (\text{Planck time}) \quad (23)$$

$$m_P = \sqrt{\frac{\hbar c}{G}} \quad (\text{Planck mass}) \quad (24)$$

$$E_P = \sqrt{\frac{\hbar c^5}{G}} \quad (\text{Planck energy}) \quad (25)$$

These quantities satisfy the fundamental relationships:

$$t_P \cdot E_P = \hbar \quad (26)$$

$$\ell_P = c \cdot t_P \quad (27)$$

$$E_P = m_P c^2 \quad (28)$$

$$\ell_P = \frac{\hbar}{m_P c} \quad (29)$$

These relations follow directly from the definitions of the Planck quantities; they are consistent with the T0 dualities and fit the view that these are anchored in the structure of spacetime.

.9 Scale Dependence and Limits of Interpretation

T0 theory shows that the different aspects of duality—time, mass, energy, length—are differently pronounced depending on the scale considered. This scale dependence is fundamental and calls for caution when interpreting extreme situations.

Complementarity of Aspects

Different aspects dominate at different scales:

- **Planck scale:** All aspects are equivalent, no approximation valid

- **Atomic scale:** Energy-time duality dominates, gravity negligible
- **Macroscopic scale:** Mass aspect dominant, quantum effects suppressed
- **Cosmological scale:** Space-time structure dominant, local quantum effects irrelevant

The Role of Small Corrections

Although the ξ parameter ($\xi = 4/3 \times 10^{-4}$) and gravitational effects are often extremely small, they still have measurable effects. These small corrections are not negligible but essential for complete understanding:

$$\text{Observable effect} = \text{Main contribution} + \xi \cdot \text{Correction} + \text{Gravitational contribution} \quad (30)$$

Caution with Singularities

Important Insight

Singularities are **not** the goal of T0 theory. They rather represent limits of applicability:

- As $r \rightarrow 0$: The local approximation breaks down
 - As $E \rightarrow \infty$: The field equations become nonlinear
 - As $T \rightarrow 0$: Time-energy duality loses its meaning
- These limits show where the theory needs to be extended.

The Complementarity Principle in T0

Analogous to Bohr's complementarity principle in quantum mechanics, T0 theory states:

$$\text{Precision}(T) \times \text{Precision}(E) \leq \text{constant} \quad (31)$$

The more precisely we determine one aspect (e.g., time), the less precise the complementary aspect (energy) becomes. This is no weakness of the theory; it is the same complementarity that quantum mechanics also knows.

Interpretation Guidelines

For correct application of T0 theory, the following guidelines apply:

1. **Scale awareness:** Always check which scale is dominant
2. **Take small effects seriously:** Don't ignore ξ corrections and gravitational effects
3. **Avoid singularities:** Understand them as hints at theoretical limits
4. **Respect complementarity:** Not all aspects can be sharp simultaneously
5. **Experimental verifiability:** Only make predictions that are measurable in principle

.10 Resolution of Quantum Paradoxes

T0 theory offers an interpretation of the classic paradoxes of quantum mechanics in which they result from an incomplete description of the underlying field structure. Once the full field dynamics is taken into account, they lose their paradoxical character in this interpretation.

Bell Correlations

T0 theory interprets the apparently instantaneous Bell correlations as follows:

- **Local condition:** $T \cdot E = 1$ at both measurement locations
- **Shared field:** Entangled particles share field configuration
- **Causal propagation:** Field changes propagate with c
- **Correlation without communication:** Pre-structured field, no signal transmission

The central thesis **[S]** is that entangled particles are not correlated through mysterious instantaneous connections, but through a shared field established when they were created. This field exists throughout the spatial region and evolves causally according to the field equations. The observed correlations result from this pre-existing field structure, not instantaneous communication.

How this interpretation relates to Bell's theorem, which rules out local models with correlations fixed at creation as long as its assumptions hold, is not worked out in this document; the thesis is therefore marked as speculative.

Wave Function Collapse

In the T0 interpretation, the collapse is only apparently instantaneous:

$$\text{Measurement} \rightarrow \text{Local field disturbance} \quad (t \sim t_{\text{Planck}}) \quad (32)$$

$$\rightarrow \text{Field propagation} \quad (v = c) \quad (33)$$

$$\rightarrow \text{Appears instantaneous since } t_{\text{Planck}} \ll t_{\text{meas}} \quad (34)$$

What appears as discontinuous collapse is, in this picture, a continuous process occurring on a time scale far below our measurement resolution. The measurement process is a local interaction between measuring device and field that creates a disturbance propagating causally.

.11 Experimental Consequences

Although most T0 effects occur on immeasurably small time scales, the theory still makes testable predictions for extreme conditions.

Prediction of Measurable Delays

For cosmic Bell tests with distance r :

$$\Delta t_{\text{measurable}} = \xi \cdot \frac{r}{c} \quad (35)$$

where $\xi = \frac{4}{3} \times 10^{-4}$ is the geometric parameter.

Numerical example:

- Satellite experiment with $r = 1000$ km:

$$\Delta t = 1.333 \times 10^{-4} \times \frac{10^6 \text{ m}}{3 \times 10^8 \text{ m/s}} \approx 0.44 \mu\text{s} \quad (36)$$

- This delay is measurable with modern atomic clocks ($\Delta t_{\text{resolution}} \sim 10^{-9}$ s)

(Note of 30 Sept. 2026: The calculation $\Delta t = 0.44 \mu\text{s}$ for 1000 km is correct, but corresponds to an influence speed $r/\Delta t = c/\xi = 7500 c$. Bell experiments already set lower bounds above this value: Salart et al., *Nature* 454, 861 (2008), about $10^4 c$; Yin et al., *Phys. Rev. Lett.* 110, 260407 (2013), $1.38 \times 10^4 c$. The prediction $\Delta t = \xi r/c$ is thus excluded in this form.)

Proposed Experiments

1. **Satellite Bell test:** Entangled photons between ground station and satellite
2. **Lunar laser ranging:** Precision measurement of quantum correlations Earth-Moon
3. **Deep space quantum network:** Test at interplanetary distances

.12 Philosophical Implications

The resolution of apparent instantaneity has consequences for our understanding of physical reality. In the T0 interpretation, nature is local and causal despite the apparent non-locality of quantum mechanics.

New Interpretation of Quantum Mechanics

T0 theory offers an alternative perspective on quantum mechanics:

New Perspective

Standard interpretation:

- Quantum mechanics requires non-locality
- Spooky action at a distance (Einstein)
- Wave function collapse

T0 interpretation:

- Everything is local in a shared field
- Correlations through field pre-structure
- Continuous, causal evolution

This perspective offers a common answer to several conceptual problems of quantum mechanics: if the apparent paradoxes result from an incomplete description, competing interpretations lose their starting point.

Unification of Quantum Mechanics and Relativity

T0 theory proposes the following resolution of the apparent conflict:

- Preserves the Lorentz invariance of field propagation
- No faster-than-light information transmission
- Quantum correlations through causal field structure

The unification is meant conceptually: both theories are understood as different aspects of the same underlying field structure. Quantum mechanics describes the coherent properties of fields, while relativity characterizes their causal structure.

.13 The Measurement Process in Detail

The measurement process in quantum mechanics has always been one of the greatest conceptual problems. Wave function collapse appears to be a non-unitary, instantaneous process fundamentally different from normal Schrödinger evolution. The T0 formalism offers an alternative description that avoids these problems.

In the T0 picture, a measurement is a local interaction between the measuring device and the field at the measurement location. This interaction occurs on the Planck time scale—extremely fast but not instantaneous. The apparent collapse is actually a very rapid but continuous reorganization of the local field structure.

Crucially, this local reorganization does not require instantaneous change of the field at distant locations. Information about the measurement propagates as a field disturbance at the speed of light. When this disturbance reaches other parts of an entangled system, it influences their further evolution, but this happens causally and at finite speed.

Within its framework, this description avoids the conceptual problems of the measurement process: it needs no separate collapse, no violation of unitarity, and no instantaneous action at a distance. Everything is described by local field interactions and causal field propagation.

.14 Quantum Entanglement Without Instantaneity

Quantum entanglement is often considered the paradigmatic example of non-local quantum phenomena. When two particles are entangled, measurement of one particle seems to instantly determine the state of the other, regardless of distance. Bell's inequalities and their experimental violation seem to prove that local realistic theories cannot reproduce quantum mechanics.

The T0 formalism offers a new perspective on these phenomena. Entanglement is not interpreted as a mysterious instantaneous connection but as the result of a shared field configuration established when the entangled particles were created. This field configuration exists throughout the spatial region between the particles and evolves according to causal field equations.

When a measurement is performed on one of the entangled particles, the measuring apparatus interacts locally with the field at that location. This interaction creates a disturbance in the field that propagates at the speed of light. The correlations between measurement results arise not from instantaneous communication but from the pre-existing structure of the shared field.

This interpretation aims to resolve the EPR paradox in a way that preserves both quantum mechanics and relativity: instead of spooky action at a distance, only local interactions with an extended field. For the open question of how this relates to Bell's theorem, see the note in the section on Bell correlations. The observed correlations result from coherent field structure, not instantaneous information transmission.